

Santosh Kushwaha

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Website: <https://santosh-kushwaha.com/>

Total Experience: 8+ years | DevOps Experience: 7+ years

DevOps | Platform Engineering | Infra Automation | Cloud & Bare-Metal | Linux

- DevOps & Infrastructure Engineer with 6+ years of experience managing large-scale Linux infrastructure spanning 50+ physical servers, with a strong focus on infrastructure automation, platform engineering, and production reliability.
- Hands-on experience in designing and operating containerized workloads, IaaS-style multi-tenant environments, and cloud-integrated systems on AWS. Proven expertise in Terraform, Ansible, Bash-driven automation, and growing experience with Go-based automation for cloud-native platforms.
- Built and maintained observability stacks using Prometheus, Grafana, Loki, and Promtail; Implemented zero-downtime deployment strategies; and automated SSL and application lifecycle management across distributed environments.
- Strong background in security hardening, monitoring, backup automation, high-availability infrastructure design, and cost-efficient operations across both cloud and bare-metal environments.

Technical Skills and Core Competencies

- **Infrastructure, Platform Engineering and Cloud**
 - Linux (Debian, Arch), Bare-metal Management (IPMI, HPE iLO)
 - Platform Engineering, Multi-Tenant Architecture, Infrastructure as a Service (IaaS)
 - Self-Hosted Platforms (MinIO, Authelia, Docker Mailserver, Monitoring Services)
 - AWS (S3, SES), GCP Storage, MinIO (S3-Compatible Object Storage), Cloudflare
- **Containers & Deployment Platforms**
 - Docker (Container Lifecycle Management, Multi-Node Environments)
 - Container Networking, Service Isolation, Resource Management
 - Kubernetes (k3s, Helm, Ingress, Services, Deployments, ConfigMaps, Health Probes, Workload Deployment)
- **Automation & Infrastructure as Code (IaC)**
 - Bash (Advanced), Python/Pytest Automation
 - Terraform (Infrastructure Provisioning)
 - Ansible (Configuration Management & Automation)
 - Go (Learning & Hands-on Projects)
 - CI/CD Automation using GitLab CI/CD
- **Networking & Web Infrastructure**
 - NGINX, Apache, Reverse Proxy, Load Balancing
 - DNS Automation, SSL/TLS Lifecycle Management, Traffic Routing
- **Monitoring & Observability**
 - Prometheus, Grafana, Loki, Promtail, Scrutiny, Uptime Kuma
 - Centralized Logging, Metrics Collection, Alerting
 - Infrastructure & HDD Health Monitoring, Incident Detection
- **Security & Access Management**
 - CrowdSec (Intrusion Prevention), Infection Monkey (Security Validation)
 - RBAC, SSH Key Management, Access Control, Security Hardening
 - Tailscale, Headscale, OpenVPN, WireGuard, Authelia (Auth & Identity Management)
- **Databases & Integrations**

- MariaDB, PostgreSQL, Redis
- Database Deployment, Configuration, Backup, and Operational Support
- Redis Configuration, Security Hardening, Socket Management, and Application Integration
- Internal Automation APIs, Custom Tooling
- Ollama, Open WebUI, LLM Deployment, Self-Hosted LLMs, OpenAI API, Claude API, AI Workflow Automation

Professional Experience

DevOps & Infrastructure Engineer

Khush Technologies, 2023 - present

- Designed and operated a multi-tenant application hosting platform spanning 50+ physical Linux servers, supporting thousands of user environments through automated provisioning, deployment, and lifecycle management workflows.
- Managed high-density containerized environments hosting networking, file management, reverse proxy, and application workloads across multi-tenant infrastructure.
- Automated infrastructure provisioning, configuration management, and deployment processes using Terraform, Ansible, Bash, and Docker, improving operational consistency and reducing manual effort.
- Built and maintained observability platforms using Prometheus, Grafana, Loki, Promtail, Scrutiny, and Uptime Kuma, improving operational visibility, incident detection, and troubleshooting efficiency.
- Implemented infrastructure security controls using CrowdSec, Infection Monkey, SSH key-based access management, Authelia, and Tailscale/Headscale to strengthen platform security and access governance.
- Designed and maintained object storage and backup workflows using MinIO, AWS S3-compatible services, and Duplicacy, supporting data protection, retention, and recovery requirements.
- Automated SSL/TLS lifecycle management, DNS provisioning, and reverse proxy configuration using Cloudflare and Route53 APIs, enabling reliable and low-risk application onboarding.
- Integrated external APIs and self-hosted LLM services using Ollama and Open WebUI into infrastructure workflows, supporting internal applications, chatbot platforms, documentation, and automated workflows.

Senior Programmer

Khush Technologies, 2021 - 2023

- Led maintenance of business-critical applications and improved application provisioning and deployment workflows, reducing manual operational effort and improving deployment consistency.
- Developed automation tools and operational utilities using Bash and Python to streamline deployments, maintenance activities, and infrastructure operations.
- Introduced containerized deployment practices by developing and maintaining Dockerfiles, including multi-stage builds, and managing container image lifecycles through GitLab Container Registry.
- Managed Linux servers, application deployments, and production troubleshooting activities across customer-facing environments.

Programmer

Khush Technologies, 2019 - 2021

- Maintained and enhanced existing PHP-based applications through bug fixes, patches, and targeted feature improvements.
- Developed application installation and provisioning workflows to streamline deployment and customer onboarding processes.
- Developed and enhanced WHMCS modules and integrations to automate provisioning workflows, streamline service management, and support hosting platform operations.
- Integrated third-party APIs and external services to extend platform functionality and automate operational workflows.
- Managed Linux-hosted application environments and developed Bash automation utilities to support deployments, troubleshooting, and operational maintenance.

Education

Bachelor of Technology (Computer Science & Engineering),
MGMCoET, Noida

First Class
Uttar Pradesh Technical University, 2008 – 2012

Languages: Hindi (Native), English (Professional)